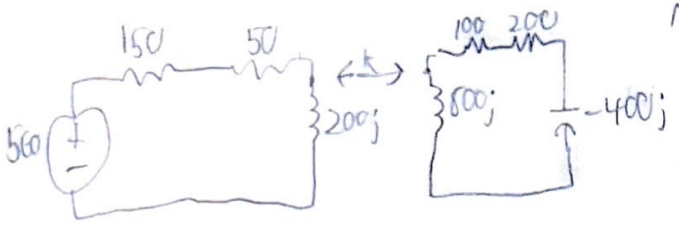
9.76 1mH ↔ 4mH

Ulan Thornberg

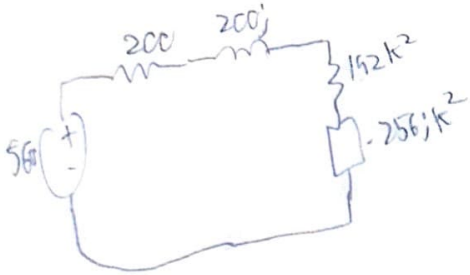


$M = k\sqrt{L_1 L_2}$

$M = k\sqrt{.00004}$

$M = .002k$

Mw



$Z_r = \frac{(200000(.002k))^2}{300 - 400j + 800j} = (192 - 256j)k^2$

$200 + 192k^2 + 200j - 256jk^2$

take deriv of mag w/ respect to k and set = 0

$Z = \sqrt{(200 + 192k^2)^2 + (200 - 256k^2)^2}$

$\frac{dZ}{dk} = \frac{1}{2\sqrt{(200+192k^2)^2 + (200-256k^2)^2}} \cdot (2(200+192k^2)384k + 2(200-256k^2)(-512k))$

$0 = 768k(200 + 192k^2) - 1024k(200 - 256k^2)$

$0 = 768(200 + 192k^2) - 1024(200 - 256k^2)$

$0 = 3(200 + 192k^2) - 4(200 - 256k^2)$

$0 = -200 + 1600k^2$

$200 = 1600k^2$

$\frac{1}{8} = k^2$

$k = \frac{\sqrt{2}}{4}$

$192(\frac{1}{8}) = 24$

$-256(\frac{1}{8}) = 32$

$200 + 24 = 224$

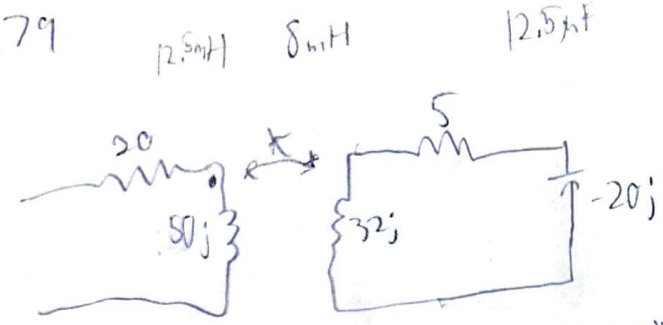
$200 - 32 = 168$

$\frac{560}{(224 + 168j)} = 1.6 - 1.2j$

$\sqrt{1.6^2 + 1.2^2} = 2$

2A

9.79



$$Z_r = \frac{(\omega M)^2}{5+12j}$$

$$Z_{eq} = 5+12j$$

$$M = k\sqrt{L_1L_2}$$

$$M = k\sqrt{1\times10^{-4}}$$

$$M = .01k$$

$$Z_r = \frac{(40k)^2}{5+12j}$$

$$Z_r = (47.34 - 113.61j)k^2$$

$$Z_{ab} = 20 + 50j + (47.34 - 113.61j)k^2$$

to be completely resistive, it must cancel out all imaginary values

$$50 = 113.61k^2$$

$$\frac{50}{113.61} = k^2$$

$$k \approx .663$$

plug in

$$Z_{ab} = 20 + 50j + (47.34 - 113.61j)(.663)^2$$

$$Z_{ab} = 20 + 50j + 20.83 - 50j$$

$$\boxed{Z_{ab} = 40.83\Omega}$$

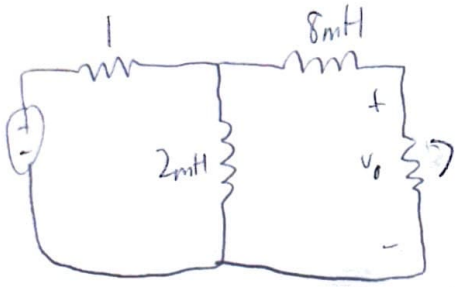
9.83

$$Z_L = 80\angle 60$$

$$Z_{new} = \frac{80\angle 60}{\left(\frac{1}{10}\right)^2} = 8000\angle 60$$

$$\frac{8000\angle 60}{\left(\frac{1}{8}\right)^2} = \boxed{512000\angle 60\,\Omega}$$

10.51



$V_{grms} = 54$

$M = k\sqrt{L_1L_2}$

$M = .75(.004)$

$M = 3mH$

voltages

$v = -54$

$R_g = i_1(1)$

$L_2 = j\omega L_2(i_1 - i_2) + j\omega M i_2 = 2j(i_1 - i_2) + 3j\, i_2$

$L_1 = j\omega L_1 i_2 + j\omega M i_1 = 8j\, i_2 + 3j(i_1 - i_2)$

$R_L = i_2(7)$

1. $-54 + i_1 + 2j i_1 + j i_2 = 0 \rightarrow i_1(1+2j) + i_2(j) = 54$

$7i_2 - 2j(i_1 - i_2) - 3j i_2 + 8j\, i_2 + 3j(i_1 - i_2) = 0 \rightarrow i_1(j) + i_2(4j + 7) = 0$

$i_1 = 12 - 21j$

$i_2 = -3$

$V_o = i_2 * R_L$

$V_o = (-3) * 7 = -21V$

$\boxed{V_{o\,RMS} = -21V}$

$|-3|^2 \cdot 7 = 63$

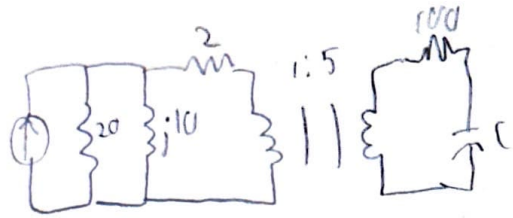
$\boxed{63 \text{ watts to } R_L}$

$54 \cdot 12 = 648$

$\frac{63}{648} = \boxed{9.7290}$

10.58

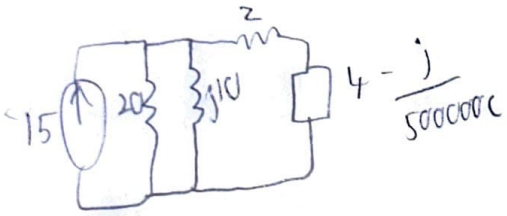
15<0

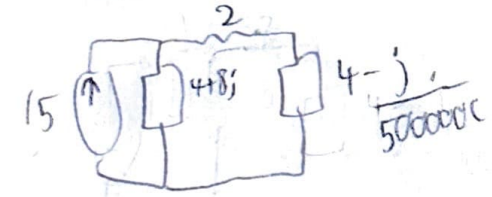


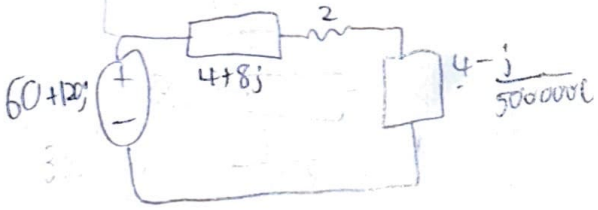
n=5

25

$$Z_r = \frac{Z_L}{n^2} = \frac{100 - \frac{j}{500000C}}{25} = 4 - \frac{j}{500000C}$$







we need this to match 8j so we can make 0 reactive power

$$6 + 8j \leftrightarrow 4 - \frac{j}{500000C}$$

$$8j \quad \frac{-j}{500000C} = 0$$

$$8 = \frac{1}{500000C}$$

$$C = \frac{1}{4000000} F$$

$$\frac{60+120j}{10} = 6+12j \leftarrow I_1$$

$$\frac{1}{5}(6+12j) = I_2$$

$$I_2 = \frac{6}{5} + \frac{12j}{5}$$

$$|I_2|^2 R =$$

$$\frac{180}{25}(100) = 720$$

for R=150

$$\frac{60+120j}{12} = 5+10j$$

$$\frac{1}{5}(5+10j) = 1+2j$$

$$|1+2j|^2 150 = 750$$

c. $\boxed{C = 250 nF}$

came from $\frac{100}{5^2}$

$6+8j$ vs $4-8j$

and thus for max power, R should equal 150 b/c

$$\frac{150}{5^2} = 6$$

$\boxed{150\Omega}$

b. $\boxed{720 \text{ watts}}$

d. $\boxed{750 \text{ watts}}$